Principles of Microeconomics

Day 4 — Duopoly: Two Firms

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Recap and today's plan

The two extremes so far

- **Perfect competition:** many firms, price $= MC$, zero profit.
- **Monopoly:** one firm, price well above MC, large profit.

Today: the middle ground

Just *two* firms — a duopoly. What happens depends on *what* they compete over: price, or quantity, and who moves first.

Learning goals

By the end you can

1. explain why two price-competing firms can end up at $P = MC$;
2. derive a firm's *response function* when firms choose quantities;
3. solve for the quantity equilibrium and compare it to monopoly and competition;
4. show why moving first can be an advantage.

One common setting throughout: demand $P = 120 - Q$, each firm's $MC = 30$.

The setting we will reuse

- Market demand: $P = 120 - Q$, where $Q = q_1 + q_2$.
- Both firms make an identical product.
- Each has constant marginal cost $MC = 30$, no fixed cost.

We will run three scenarios on this same setting so the numbers are directly comparable: price competition, quantity competition, and a first-mover.

Competing by setting prices

Suppose each firm posts a *price*, and buyers flock to the cheaper one.

The pressure to undercut

- If your rival charges above MC, you can undercut slightly and take the *whole* market.
- Your rival then undercuts you back.
- The product is identical, so buyers always chase the lower price.

Where the price war ends

The stopping point

Undercutting only stops when neither firm can profitably cut further — that is at

$$P = MC = 30.$$

- At $P = 30$: total quantity $Q = 120 - 30 = 90$.
- Each firm earns *zero* profit.
- Remarkable: just *two* firms reproduce the competitive outcome.

Why price competition is so fierce

The intuition to hold on to

With identical products and price-setting, there is no reward for being “almost” the cheapest — you need to be *the* cheapest.

- That all-or-nothing pull drives the price straight down to cost.
- It is a useful benchmark, but a stark one — most real duopolies are gentler.
- That is why we now turn to competition over *quantities*.

Competing by choosing quantities

Now suppose each firm instead picks a *quantity* to produce, and the market price is whatever clears total output.

The new tension

- Each firm's output affects the price faced by *both*.
- So each firm must decide while taking the rival's output as given.

This leads to the idea of a *response function*.

Firm 1's problem

Firm 1 treats firm 2's output q_2 as fixed and maximises its own profit:

$$\pi_1 = \big(P - MC\big)q_1 = \big(120 - q_1 - q_2 - 30\big)q_1 = (90 - q_1 - q_2)\, q_1.$$

- Expand: $\pi_1 = 90q_1 - q_1^2 - q_1 q_2$.
- We now find the q_1 that makes this as large as possible, for a given q_2.

Firm 1's response function

Take the derivative with respect to q_1 and set it to zero:

$$\frac{d\pi_1}{dq_1} = 90 - 2q_1 - q_2 = 0 \quad \Longrightarrow \quad q_1 = 45 - \tfrac{1}{2}q_2.$$

- This is firm 1's *best output for each level of* q_2.
- It slopes down: the more the rival makes, the less firm 1 wants to make.

Both response functions

The two firms are symmetric, so firm 2's response mirrors firm 1's:

$$q_1 = 45 - \tfrac{1}{2}q_2, \qquad q_2 = 45 - \tfrac{1}{2}q_1.$$

- Equilibrium is where *both* hold at once — each firm's output is a best reply to the other's.
- Graphically, that is where the two lines cross.

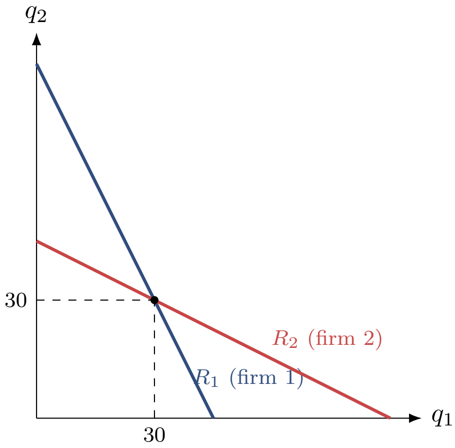
Solving the two equations

Substitute the second into the first:

$$
\begin{aligned}
q_1 &= 45 - \tfrac{1}{2}\left(45 - \tfrac{1}{2}q_1\right) = 45 - 22.5 + \tfrac{1}{4}q_1 \\
\tfrac{3}{4}q_1 &= 22.5 \quad \Longrightarrow \quad q_1 = 30.
\end{aligned}
$$

By symmetry $q_2 = 30$ as well. So each firm produces 30.

The quantity equilibrium, drawn



Reading the response-function diagram

- Each line is one firm's best output as a function of the rival's.
- They slope down: when one firm makes more, the other's best reply is to make less.
- The crossing point $(30, 30)$ is the only pair where *both* firms are simultaneously happy.
- Away from it, at least one firm would want to change its output.

The equilibrium numbers

With $q_1 = q_2 = 30$:

- Total output $Q = 60$.
- Price $P = 120 - 60 = 60$.
- Each firm's profit $= (60 - 30) \times 30 = 900$.
- Industry profit $= 1800$.

Where does the duopoly sit?

Compare the three structures on the same setting:

Structure	Total Q	Price P	Industry profit
Monopoly (one firm)	45	75	2025
Quantity duopoly	60	60	1800
Perfect competition	90	30	0

Two firms competing on quantity land *between* monopoly and competition — more output and a lower price than monopoly, but not all the way to $P = MC$.

What if one firm moves first?

Now let firm 1 (the *leader*) commit to its output *before* firm 2 (the *follower*) chooses.

The leader's advantage

The leader knows the follower will react along its response function $q_2 = 45 - \frac{1}{2}q_1$, and can bake that reaction into its own decision.

The leader's calculation

Substitute the follower's response into the price the leader faces:

$$P = 120 - q_1 - q_2 = 120 - q_1 - \left(45 - \tfrac{1}{2}q_1\right) = 75 - \tfrac{1}{2}q_1.$$
$$\pi_1 = (P - 30)q_1 = \left(45 - \tfrac{1}{2}q_1\right)q_1.$$

Maximise: $\dfrac{d\pi_1}{dq_1} = 45 - q_1 = 0 \Rightarrow q_1 = 45.$

The first-mover outcome

- Leader: $q_1 = 45$. Follower: $q_2 = 45 - \frac{1}{2}(45) = 22.5$.
- Total $Q = 67.5$, price $P = 120 - 67.5 = 52.5$.
- Leader profit $= (52.5 - 30)(45) = 1012.5$.
- Follower profit $= (52.5 - 30)(22.5) = 506.25$.

The lesson

By committing first, the leader produces more, earns *twice* the follower's profit — a genuine first-mover advantage.

All four outcomes together

Structure	Total Q	Price P	Industry profit
Monopoly	45	75	2025
Quantity duopoly	60	60	1800
First-mover duopoly	67.5	52.5	1518.75
Perfect comp. / prices	90	30	0

As we move down the table: more output, lower price, less total profit. More competition always helps buyers.

Day 4 — checkpoint

You should now be able to

- explain why price competition with identical goods drives P to MC;
- derive a firm's response function from its profit;
- solve for the quantity equilibrium where both responses agree;
- rank monopoly, duopoly and competition by output and price;
- explain the first-mover advantage.

Week 2 — the big picture

One thread through four days

- **Day 1:** the cost curves — the firm's technology.
- **Day 2:** competition — price taking, $P = MC$, zero long-run profit.
- **Day 3:** monopoly — $MR = MC$, a price above cost, deadweight loss.
- **Day 4:** duopoly — outcomes sitting between the two extremes.

The number of firms, and how they compete, decides how much output reaches buyers and at what price.

In-class exercise

Exercise 4 — Duopoly

Work through **Exercise 4** on the sheet: a fresh demand curve and cost. You will find the price-competition outcome, derive the response functions and solve the quantity equilibrium, work out the first-mover case, and tabulate all four structures.